RESULTS AT A GLANCE

More reliable rowing measurements

The strongest gains are steadier seat tracking, better selection of complete body poses, and more defensible stroke timing. The app's UI and offering were not changed.

96.0%
tracking coverage after

+619
observed seat-frames

808
regression tests run

What changed overnight

Tracking stays with the rower
An ordered seat tracker uses hip position and shared boat/camera movement to reduce missing tracks and seat mix-ups.

Body poses survive crop boundaries
A complete, stronger torso estimate can replace a truncated duplicate. This fixes a reproduced hip/knee collapse in clear footage.

Motion and timing use stronger evidence
Confidence-aware body filtering, consistent fractional event times, supported hand-based finishes, and one-use crew catch matching improve the analysis core.

Measured scope. Coverage rose from 90.2% to 96.0% on five cached boat segments from three supplied videos. This is tracking availability, not a claim that 96% of joints or strokes are correct. [1]

Status at handoff: changes and reproducible evaluations are saved locally. Fresh inference-to-report runs completed. Neural weights were not retrained; the work improves detection postprocessing, tracking and analysis around the existing pose model.

MEASURED TRACKING GAINS

Fewer gaps; more stable seat identity

The same cached detections were fed to the original and revised tracking paths, isolating tracking changes from new inference. [1]

Footage segment	Before	After	Change
0185 / boat 1	62.4%	96.4%	+34.1 pp
0185 / boat 2	92.0%	94.2%	+2.2 pp
9673 / boat 1	91.6%	94.1%	+2.5 pp
9674 / boat 1	97.5%	97.7%	+0.2 pp
9674 / boat 2	99.9%	98.2%	-1.7 pp
All five segments	**90.2%**	**96.0%**	**+5.8 pp**

The tracker retained **10,272 of 10,704** possible seat-frame observations, up from **9,653**. Missing observations fell from 1,051 to 432, a **59% reduction**. No predicted position is counted as a detected observation.

Small visual identity audit

Across 20 frames / 160 seat checks	Before	After
Correctly assigned	141	153
Wrong-seat assignments	3	0
Missing assignments	16	7

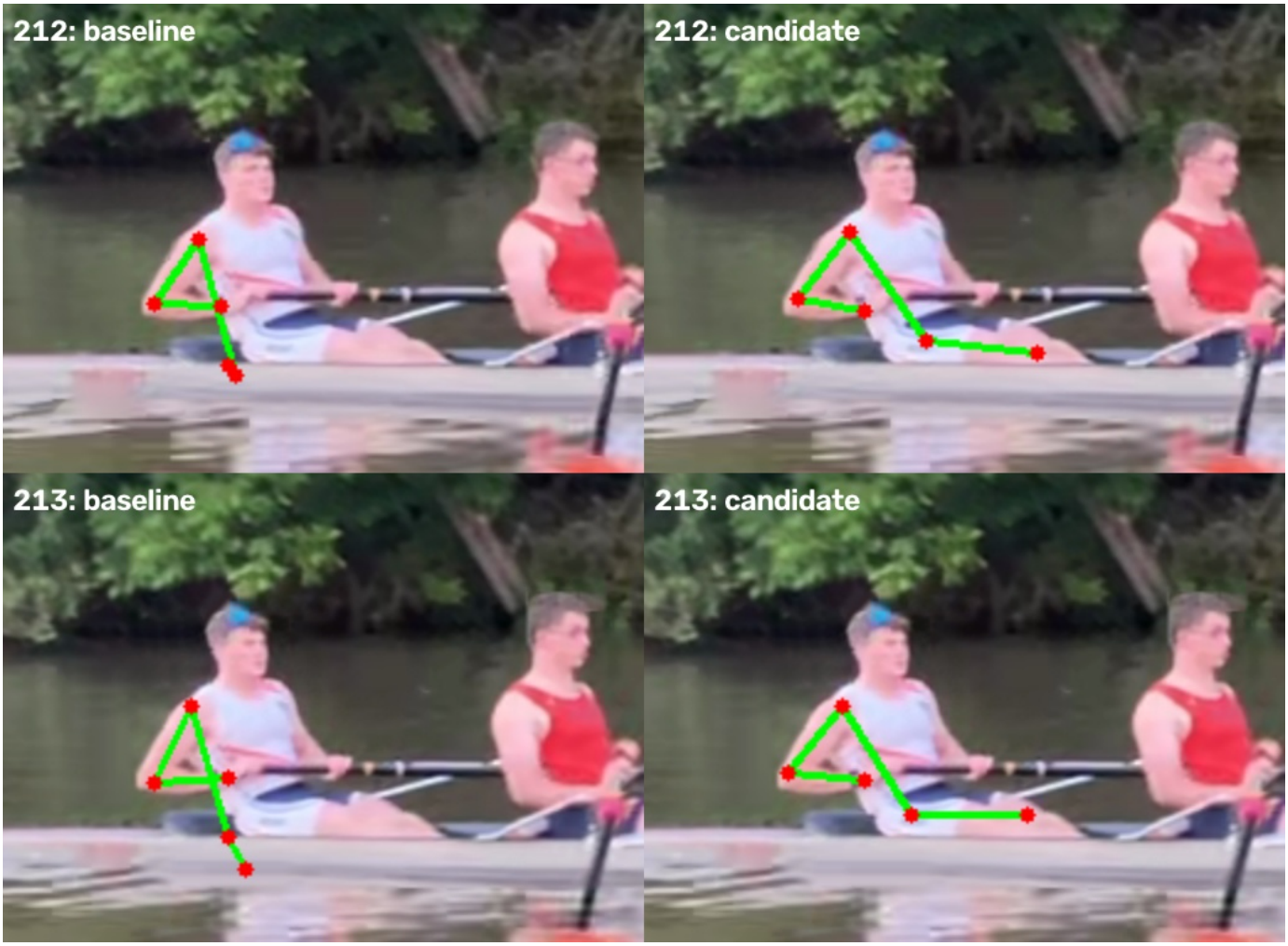
Tradeoffs and limits. Improvements were not uniform. The second 9674 segment lost 1.7 percentage points of coverage. In the first 9674 visual subset, two previously correct observations became missing. The audit was assistant-reviewed, selected for development, and not independently labeled.

pp means percentage points; changes use unrounded values. The largest gain came on the difficult, distant 0185 first-boat segment: 62.4% to 96.4%. Cleaner footage had less room to improve. These segment results predate the final crop-boundary detector fix; they should not be presented as a fresh benchmark of the entire final pipeline.

BODY POSE CORRECTION

A complete body beats a cropped torso

A reproduced failure showed that person-detection confidence could favor a crop cutting through the rower over a more complete body estimate. [2]



Left: previous duplicate selection. **Right:** revised selection on the same raw proposals. Red points and green links show the selected pose. The complete crop restores hip and knee locations to the visible body. Two consecutive sampled frames from IMG_9674 are shown.

The revised rule acts only after an anatomical same-person match. A complete alternative must have comparable person confidence and materially stronger shoulder/hip confidence. It keeps one whole pose and retains partial-only detections when no suitable replacement exists.

Observed result. The two selected torso estimates changed from about -15 degrees to -36/-33 degrees, removing a crop-induced jump of roughly 17-20 degrees. This is a targeted failure correction, not a measured reduction in general angle error. Detection counts stayed unchanged across 12 replayed frames; six existing arm-audit predictions were unchanged.

MOTION AND ROWING ANALYSIS

More precise signals and phase timing

The core now handles confidence, gaps and event clocks more consistently. The numerical gains below come from controlled tests, not independent real-video labels.

Body-motion conditioning

Body shape is filtered relative to the hip, so shared camera movement does not become artificial joint movement. A fixed anatomical side avoids frame-by-frame left/right splicing. Isolated, well-supported joint jumps are rejected; interpolated gaps remain explicitly marked as estimates.

Controlled noisy-motion test [3]	Before RMSE	After RMSE
Trunk angle	3.29 degrees	2.14 degrees
Elbow angle	8.72 degrees	5.75 degrees
Thigh angle	3.95 degrees	2.48 degrees
Hand position, torso lengths	0.0549	0.0349

RMSE is root-mean-square error; lower is better. These rows average six rate/frame-rate settings with confidence-dependent simulated noise. The full benchmark contains 24 conditions. Clean trunk motion was unchanged in that benchmark; simulated confidence/noise correlation is an assumption.

Stroke and crew timing

One event clock. Rate, drive and recovery durations now use consistent fractional catch/finish times. Ideal-signal rate error from frame quantization fell from 0.425 to 0.00171 spm across 12 cases. This tests numerical precision only. [4]

Finish follows supported hand motion. Knee extension can remain flat into recovery. Where local hand evidence is sufficient, hand reversal determines finish. Source review found and corrected gross late finishes on two videos; flat or blurred hand motion still limits exact timing. [5]

Missing data and synchrony. Rhythm preserves the real sample clock instead of compressing gaps. Crew catches require a strict majority of distinct identities; each catch is used once. Insufficient support clears or withholds the timing score. [6]

Coincident or nonfinite joint endpoints now yield unavailable angles rather than fabricated 90-degree or upright measurements. No ideal rowing form or fixed limb length is imposed.

VERIFICATION AND RESTRAINT

Validation and experiment results

Changes were checked with targeted regressions, controlled motion, cached-footage comparisons, source-frame review and fresh full-pipeline runs.

Validation at the pause point

808 tests ran: 798 passed, 10 failed. The ten failures match pre-existing application/job/upload/web issues; no new failures were found. This is not an all-green suite. All four standalone legacy rowing/fault checks also passed. [7]

Fresh pipeline run	Output diagnostics	Result
IMG_9673 / 20 seconds	8 rowers; 36 cycles; 6 crew events	Completed; checks passed
IMG_9674 / 21 seconds (final crop-boundary fix)	8 rowers; 54 cycles; 9 crew events	Completed; checks passed

Saved-output checks cover timing arithmetic, observation counts, unique crew-event use and evidence integrity. Counts and estimated rates are diagnostics, not accuracy labels. The 9673 run preceded the final crop fix; the 9674 run included it. [8]

Experiments not adopted

Experiment	Why it stayed out of production
Larger pose backbones	No convincing localization win in the small audit; medium inference cost was about 5x the nano model in the controlled CPU comparison.
Tighter person crops	Mixed joint-placement results; no consistent improvement.
Neighbor-frame optical flow and mirrored inference	No convincing aggregate gain; some wrist estimates worsened.
Every source frame / wider smoothing	Full-rate processing cost more without established real-video accuracy gain. Wider filtering reduced random noise but blurred clean motion and barely helped persistent errors.

These decisions preserve useful detail and avoid adding complexity or inference cost on the strength of smoother-looking output alone. The original nano model remains the default. [3, 9]

HANDOFF AND EVIDENCE

What is stronger - and what remains

The work is paused. This report summarizes the saved implementation and evidence; it does not claim the model is finished or independently validated.

Remaining precision limits

Wrist and hip localization can still be wrong despite high detector confidence. Some finishes span several similar poses, so an exact timestamp is not visually resolvable. Angles are projected 2D measurements and remain sensitive to viewpoint, overlap and hidden anatomy. The new crop rule has targeted regression evidence, not broad held-out validation.

Recommended next work when resumed

1. Build an independently labeled rowing benchmark covering clear side-on motion, different stroke rates, body types and cameras.
2. Expand crop-boundary and phase-selection comparisons on footage not used to tune the changes.
3. Use those labels to guide pose-model refinement or fine-tuning, with joint-position and event-time error as the acceptance criteria.

Evidence ledger

Paths below are relative to **app/strokesense/** in the local workspace. Frozen baselines, replay scripts and detailed caveats are retained alongside the results.

[1] Tracking - evaluation/tracking/results.json; README.md

[2] Crop correction - evaluation/body_motion/README.md; crop_edge_compare.py

[3] Body-motion tests - evaluation/body_motion/synthetic-comparison.json

[4] Fractional timing - evaluation/stroke_timing/README.md; results.json

[5] Event source review - evaluation/events/README.md; comparison.json

[6] Rhythm and crew timing - evaluation/rhythm/README.md; evaluation/synchrony/README.md

[7] Final regression log - results/body-motion-evaluation/crop-edge-comparison/full-tests.log

[8] Fresh pipeline checks - evaluation/pipeline/results.json; results/crop-edge-pipeline/verification.json

[9] Sampling experiments - evaluation/sampling/README.md; filter_support_results.json

Scope preserved. No UI or offering changes, no neural-weight retraining and no footage upload as part of this work. Existing workspace changes were preserved. The full work record is evaluation/ACCURACY_WORK_LOG.md.